Blood Cells Classification

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Abstract

Microscopic examination of peripheral blood smears is a fundamental diagnostic procedure in hematology laboratories. However, manual classification of blood cells is time-consuming and subject to interobserver variability.
This project presents a machine learning pipeline for the automated classification of eight normal blood cell types using smear images. Classical machine learning models, including Random Forest and Support Vector Machine (SVM), were implemented and optimized using hyperparameter tuning.
In addition, a convolutional Autoencoder was adopted as a deep learning approach and trained exclusively to model normal blood cell morphology and enable anomaly detection through reconstruction error.

1 Introduction

Blood smear examination plays a central role in diagnosing hematological diseases. Blood is composed of several cellular components including red blood cells (erythrocytes), white blood cells (leukocytes), and platelets. Blood cells exist in several subtypes, each characterized by distinct morphological features such as nuclear structure, cytoplasmic texture, granularity, and staining intensity. The main subtypes include:

- Neutrophils
- Lymphocytes
- Monocytes
- Eosinophils
- Basophils
- Immature granulocytes
- Erythroblasts
- Platelets

The identification and classification of these cell types are essential for differential blood counts. Abnormal distributions of these cells may indicate conditions such as infections, inflammatory disorders, or hematological malignancies including leukemia. However, manual microscopic inspection requires skilled personnel and may introduce subjectivity. With recent advances in researches, automated image-based classification systems have become feasible and reliable where machine learning algorithms can analyze morphological features extracted from blood smear images and classify cell types with high accuracy. Furthermore, deep learning models have demonstrated remarkable success in medical image analysis by learning complex visual patterns directly from raw image data. This project aims to explore the use of machine learning techniques for multi-class classification of blood cell images and evaluate their performance. In addition, a deep learning-based Autoencoder model is investigated for modeling normal cell morphology and detecting abnormal cells.

2 Dataset Description

This project integrates two complementary blood cell image datasets that together provide comprehensive coverage of both normal peripheral blood morphology and pathological cells associated with acute myeloid leukemia (AML).

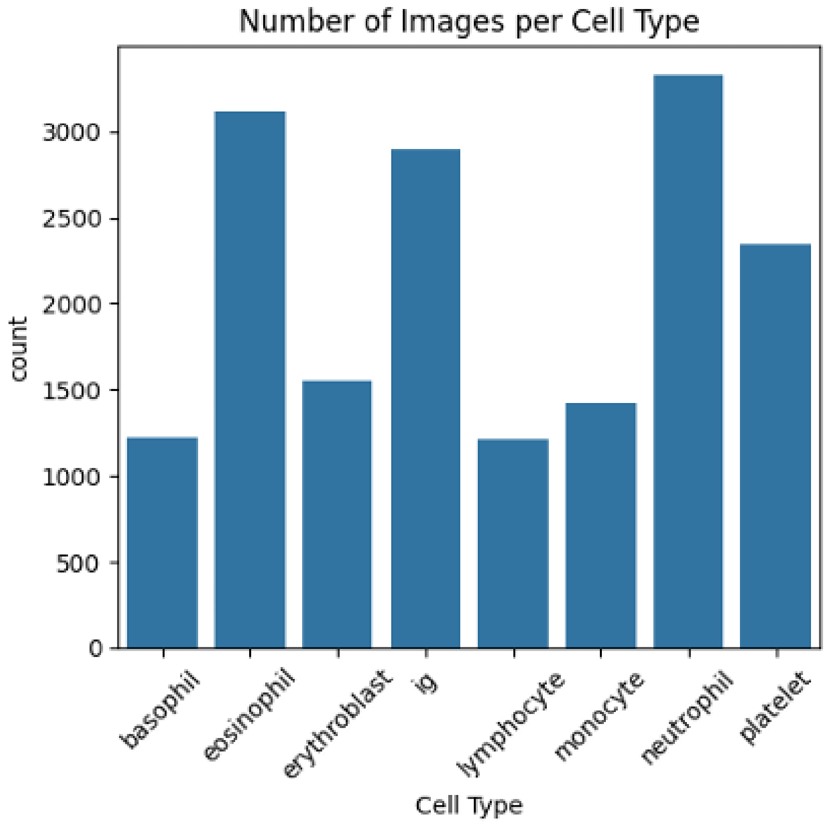
The Mendeley dataset supplies robust representation of normal cell types essential for differential diagnosis and the TCIA AML dataset contributes pathological blast populations and rare immature precursors critical for leukemia detection.

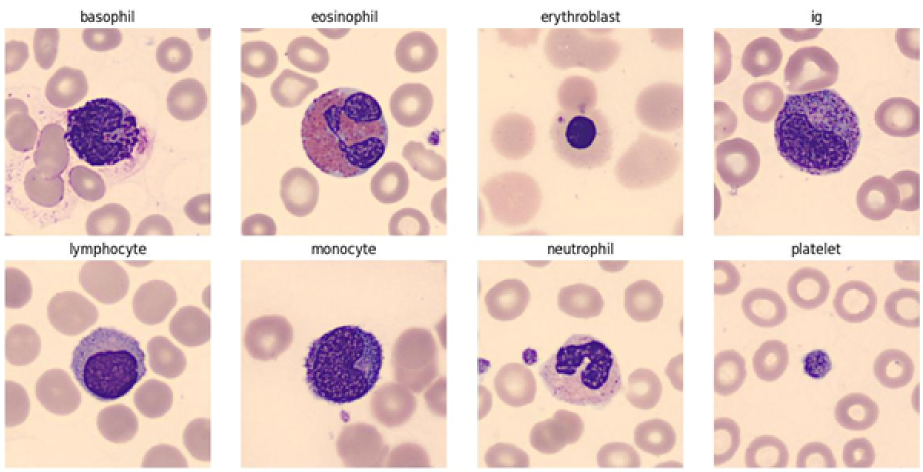
2.1 Mendeley Blood Cell Dataset

The first dataset comprises 17,092 microscopic images of normal peripheral blood cells obtained from the Hospital Clínic de Barcelona's Core Laboratory [Acevedo et al., 2019]. Images were captured using the CellaVision DM96 digital hematology analyzer at a resolution of 360 × 363 pixels in RGB JPG format. Blood samples were collected from donors without infections, hematologic or oncologic diseases, and pharmaceutical treatments at the time of acquisition. The smears were stained using the standardized May-Grünwald Giemsa protocol, ensuring consistent morphological visualization across samples. The dataset encompasses eight morphological classes including normal leukocyte populations and precursors:

- Neutrophils (segmented and band forms)
- Eosinophils
- Immature granulocytes (metamyelocytes, myelocytes, and promyelocytes)
- Platelet
- Erythroblasts
- Monocytes
- Basophils
- Lymphocytes (typical)

Index	Cell Type	count	percentage
0	Neutrophil	3329	19.48
1	Eosinophil	3117	18.24
2	ig	2896	16.94
3	Platelet	2348	13.74
4	Erythroblast	1551	9.07
5	Monocyte	1420	8.31
6	Basophil	1218	7.13
7	Lymphocyte	1214	7.10



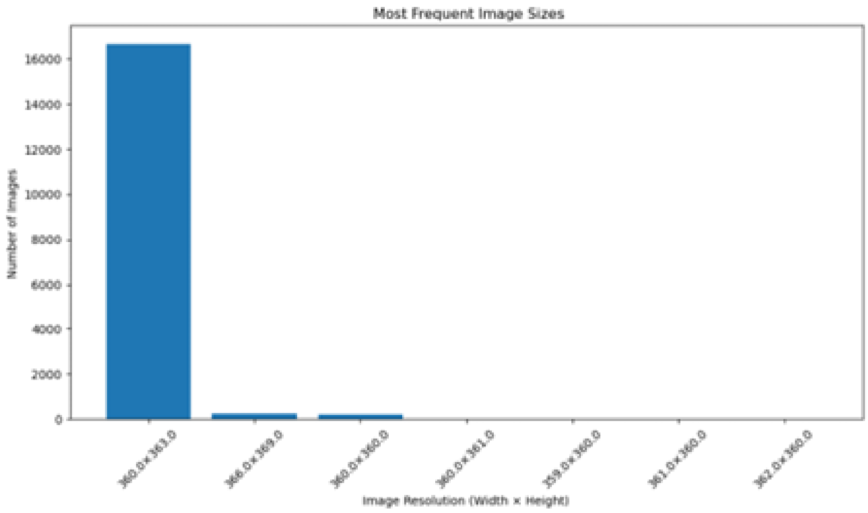


2.1.1 Image Resolution Distribution

The analysis of image resolutions shows that the dataset contains a variety of image sizes. Where the images have many different width–height combinations and most are around 360×360, but not all. This variety in images sizes means that some pre-processing is necessary. Therefore, it is recommended to resize all images to a standard resolution while keeping their aspect ratios, to maintain consistency and avoid any distortion. Moreover, considering the image resolution statistics:

- Mean width: 360.09 pixels
- Mean height: 363.05 pixels
- Standard deviation of width: 0.72 pixels
- Standard deviation of height: 0.79 pixels

We confirm that the average image resolution was found to be 360.09 × 363.05 pixels. The standard deviations of width and height were 0.72 pixels and 0.79 pixels, respectively, indicating extremely low variability in image dimensions. This suggests that the images are nearly uniform in size. Consequently, resizing all images to a fixed resolution is statistically justified and is unlikely to introduce geometric distortion or loss of relevant cellular features.



Width	Height	Count
360.0	363.0	16639
366.0	369.0	250
360.0	360.0	198
360.0	361.0	2
359.0	360.0	1
361.0	360.0	1
362.0	360.0	1

2.1.2 Data Quality Assessment

The dataset is arranged so that each folder represents a different types of blood cells and that all images inside a folder share the same label. This structure ensures that the dataset is fully labeled and suitable for supervised learning. A detailed data quality check was carried out, leading to the following observations:

- No missing or incorrect labels were found.
- All image files paths were valid and accessible.
- No corrupted image files were detected.

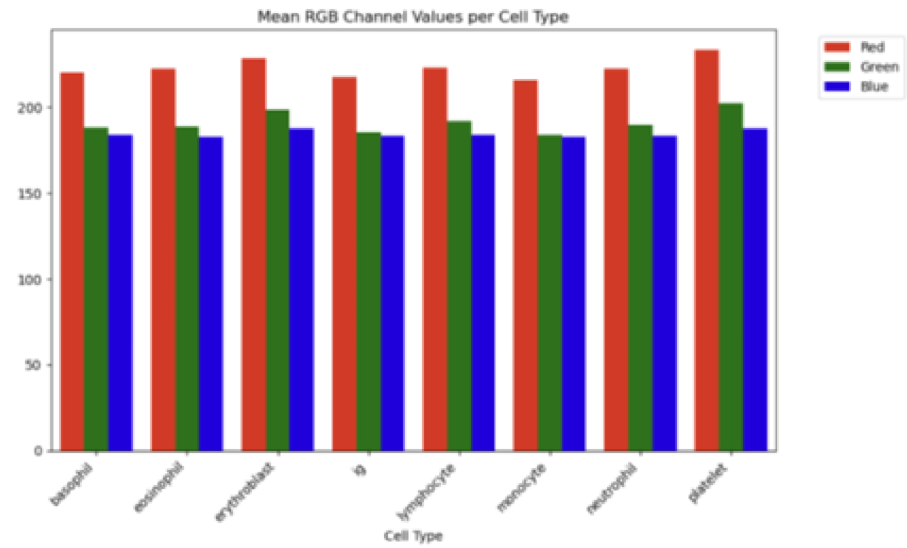
- Only one image could not be read during resolution analysis and was removed from the dataset.

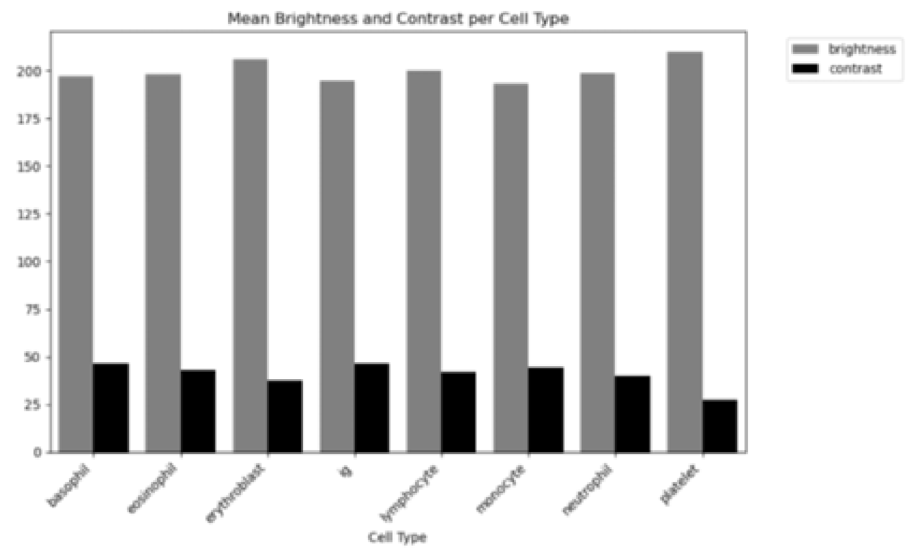
Overall, the dataset is clean and well-organized.

2.1.3 RGB and Grayscale Intensity Analysis

To understand the characteristics of different cell types in the datasets, we perform fundamental image analysis based on intensity and color distributions. Grayscale analysis provides insights into overall brightness and contrast patterns by converting RGB images to luminance values using the standard ITU-R BT.601 weighting (0.299R + 0.587G + 0.114B).

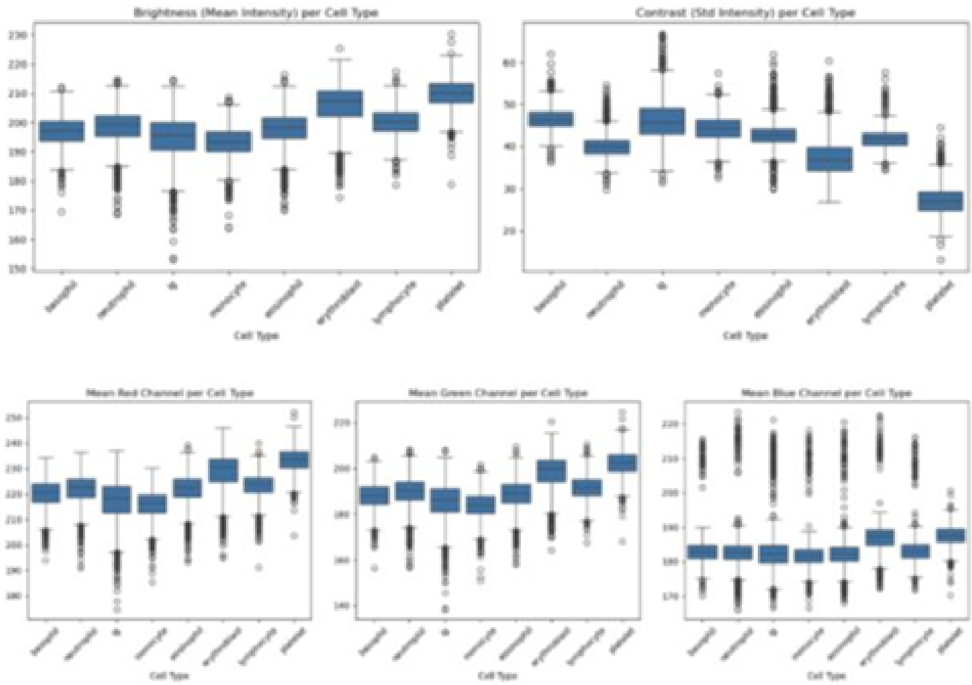
Channel-specific RGB statistics reveal color variations that may be associated with different staining properties or morphological features of cell types. These basic metrics serve as a foundation for quality control, helping to identify potential imaging inconsistencies, batch effects or significant distributional differences between classes.





The mean RGB channel values reveal relatively consistent color profiles, with all classes showing red channel dominance (approximately 215-235) followed by green (185-202) and blue (183-190) channels. Erythroblast and Platelet exhibit the highest red channel intensities (232 and 235 respectively), while ig displays the lowest overall intensity across all channels. The grayscale analysis demonstrates relatively uniform brightness levels (193 209) across most cell types. Platelet shows the highest mean brightness (209) with distinctly lower contrast (27), while ig, monocyte and basophil show slightly reduced brightness (195-197) with higher contrast values (47-48).

2.1.4 Distribution Analysis with Boxplots



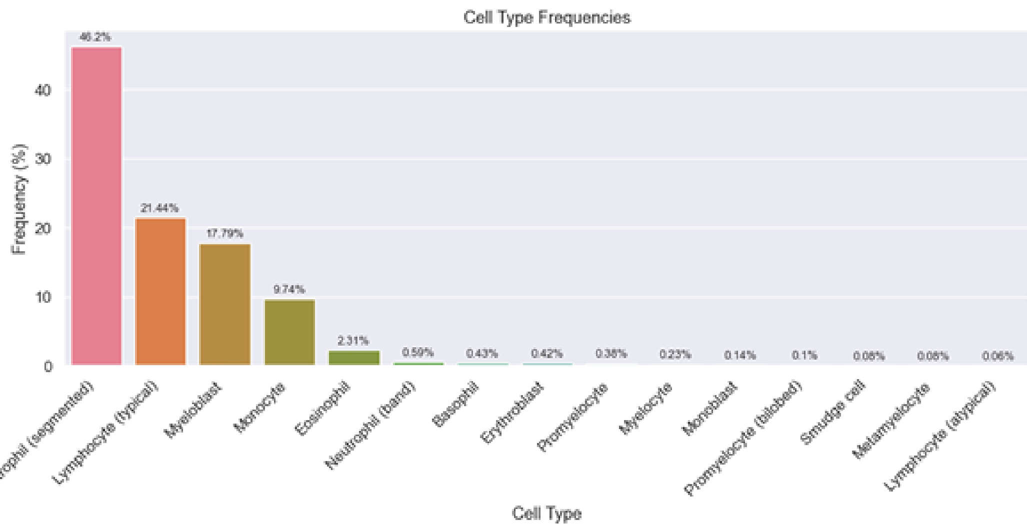
The boxplot visualizations show that most cell types display relatively compact interquartile ranges with medians around 195-210, though all classes exhibit outliers. Ig displays numerous low-intensity outliers extending down to 155 and exceptionally high contrast variability with outliers extending above 60. Platelet demonstrates the narrowest contrast distribution with a median around 27, confirming its more homogeneous internal structure. For the RGB channels, ig exhibits numerous outliers spanning from 175 to above 250 in the red channel, while platelet shows elevated green median values (202). The prevalence of outliers across all metrics suggests some degree of imaging variability that could potentially be addressed through outlier filtering or normalization strategies during preprocessing.

2.2 The Cancer Imaging Archive (TCIA) AML Dataset

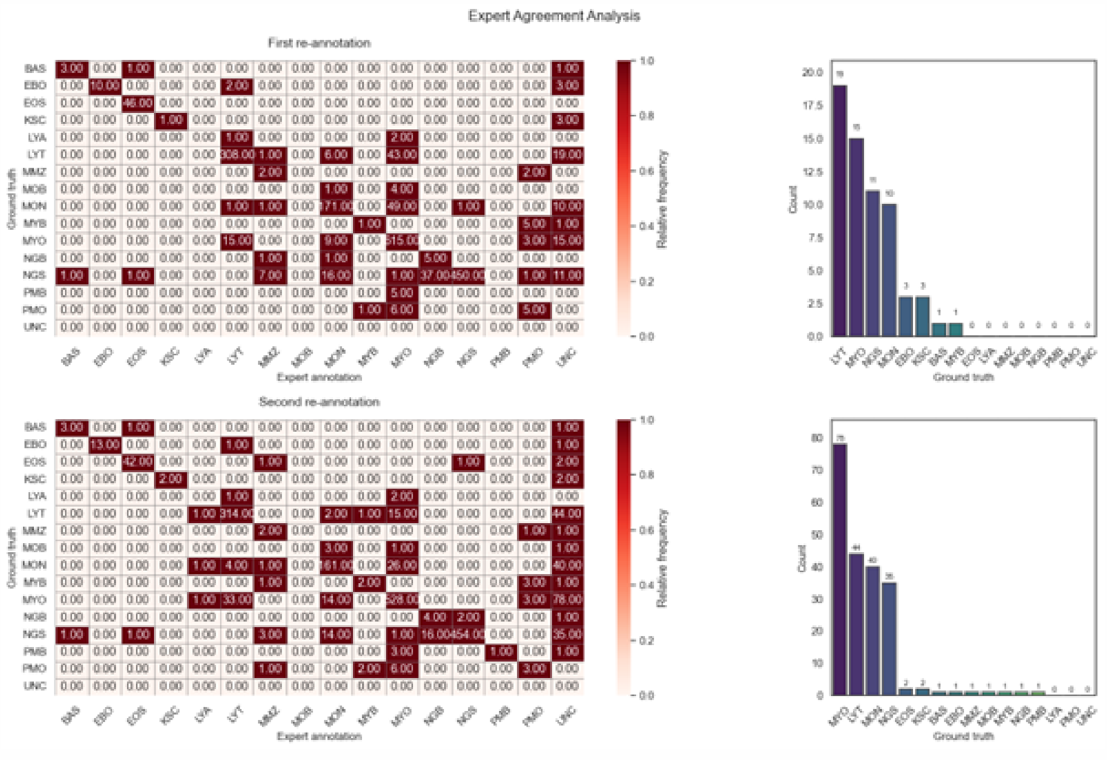
The second dataset contains 18,365 single-cell images extracted from peripheral blood smears of 200 individuals: 100 patients diagnosed with various subtypes of acute myeloid leukemia (AML) and 100 patients exhibiting no morphological features of hematological malignancies [Matek et al., 2019].

Each cropped image represents a single white blood cell and was assigned a label according to clinically relevant cytomorphological categories used in hematology diagnostics. The dataset includes mature leukocytes, immature granulocyte precursors and pathological cell types such as myeloblasts, which are crucial for AML diagnosis. The images reflect real clinical variability, including staining differences, background artifacts and class imbalance, making the dataset particularly suitable for developing robust computer vision models for blood cell classification and leukemia screening tasks.

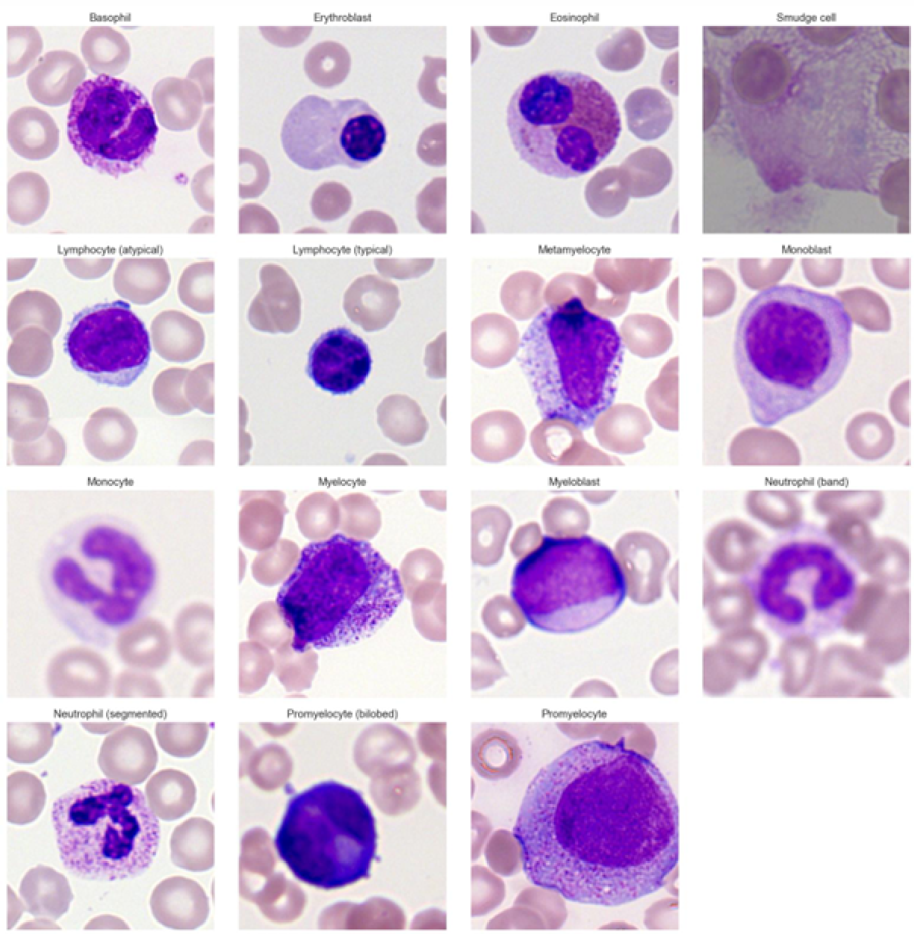
index	cell_type	count	frequency
0	Neutrophil (segmented)	8484	46.20
1	Lymphocyte (typical)	3937	21.44
2	Myeloblast	3268	17.79
3	Monocyte	1789	9.74
4	Eosinophil	424	2.31
5	Neutrophil (band)	109	0.59
6	Basophil	79	0.43
7	Erythroblast	78	0.42
8	Promyelocyte	70	0.38
9	Myelocyte	42	0.23
10	Monoblast	26	0.14
11	Promyelocyte (bilobed)	18	0.10
12	Smudge cell	15	0.08
13	Metamyelocyte	15	0.08
14	Lymphocyte (atypical)	11	0.06



A subset of 1,905 single-cell images was re-annotated by a same examiner (using only cropped cell images, without smear context). The re-annotation was performed twice, with an 11 months gap, allowing assessment of both inter- and intra-examiner agreement. The examiner could label images as “unclear” (63 images in the first pass, 208 in the second), reflecting uncertainty in some cases. For clearly assigned labels, agreement with the gold standard was excellent (Cohen’s $=0.84$ and 0.87). thus to mention that some classification challenges are inherent to human cytomorphology rather than model limitations.



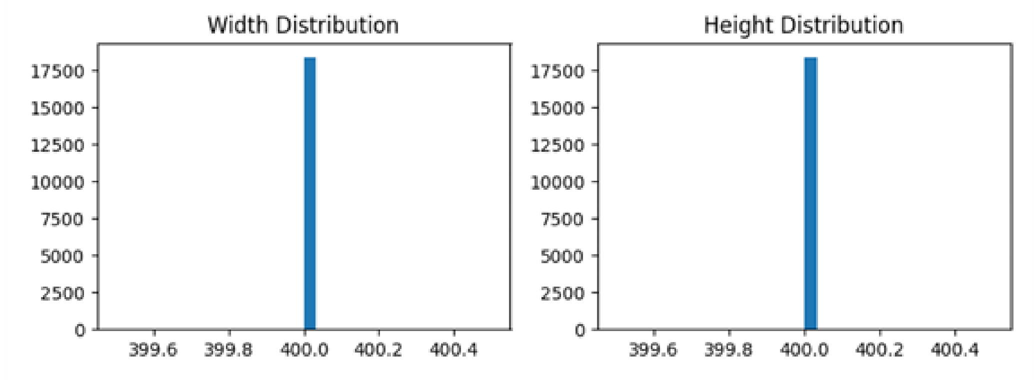
Expert Agreement Analysis
First re-annotation
Second re-annotation
Ground truth
Expert annotation
Relative frequency
Count
BAS
EBO
EOS
KSC
LYA
LYT
MMZ
MOB
MON
MYB
MYO
NGB
NGS
PMB
PMO
UNC



Basophil
Erythroblast
Eosinophil
Smudge cell
Lymphocyte (atypical)
Lymphocyte (typical)
Metamyelocyte
Monoblast
Monocyte
Myelocyte
Myeloblast
Neutrophil (band)
Neutrophil (segmented)
Promyelocyte (bilobed)
Promyelocyte

2.2.1 Image Resolution Distribution

The image resolution analysis shows that all leukemia cell images have a uniform resolution of 400 × 400 pixels and TIFF format, with a total of 18,365 images sharing the same dimensions. This high level of consistency indicates that the dataset was created under standardized imaging conditions.
Because all images have identical resolutions, no additional resizing is required to harmonize image sizes across the dataset. This simplifies the preprocessing pipeline and helps prevent potential information loss or distortions caused by rescaling. Overall, the uniform resolution contributes to a cleaner and more reliable dataset for blood cell classification tasks.

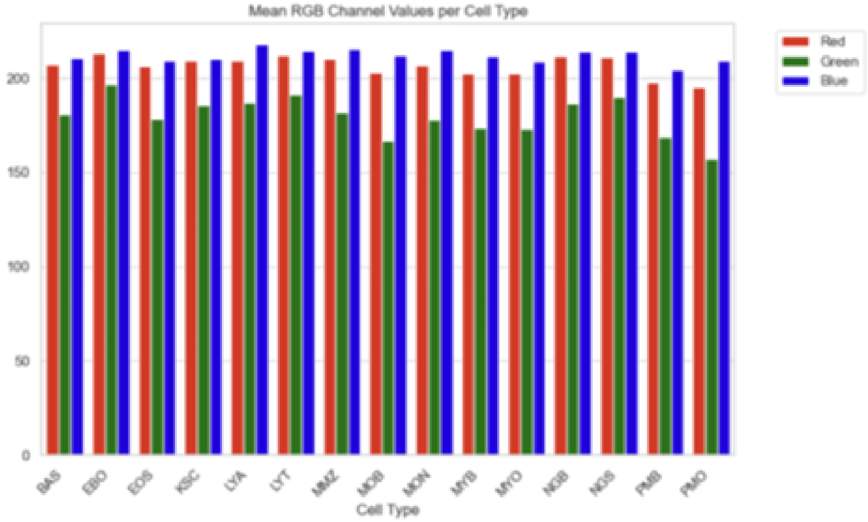


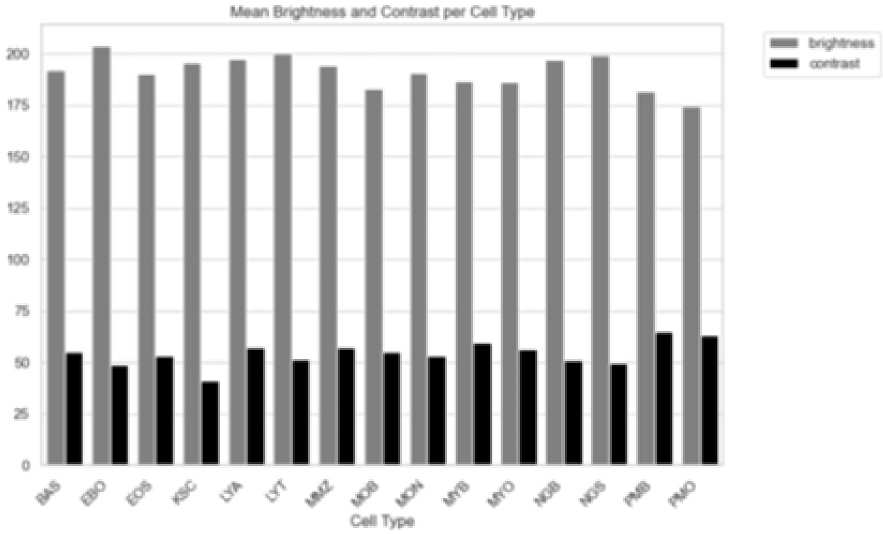
2.2.2 Data Quality Assessment

A data quality analysis was conducted on the leukemia dataset to verify image integrity and label consistency. In total, 18,365 images were analyzed and all images were successfully loaded without errors. No corrupted or unreadable image files were detected during the analysis.
All image file paths were valid, accessible and no missing labels were found. The dataset contains 15 distinct leukemia-related blood cell types, represented by consistent class labels across the folder structure.
Overall, the leukemia dataset is clean, well-organized, and fully labeled, making it suitable for supervised learning and further blood cell classification tasks.

2.2.3 RGB and Grayscale Intensity Analysis

The mean RGB channel values show consistent patterns, with red values clustering around 200-210, green at 155-195, and blue at 205-220. The red channel dominance observed in the first dataset is less pronounced here, with blue channel intensities elevated. The grayscale brightness analysis reveals moderate variation (175-205) across cell types. PMO shows notably lower brightness values (¡ 175), while EBO exhibits higher brightness (¿ 200). Contrast values are relatively uniform (approximately 40-60), with KSC displaying the lowest contrast (40) and PMO and PMB showing elevated levels (65). Overall, the dataset displays greater variability in brightness and color profiles compared to the first dataset.





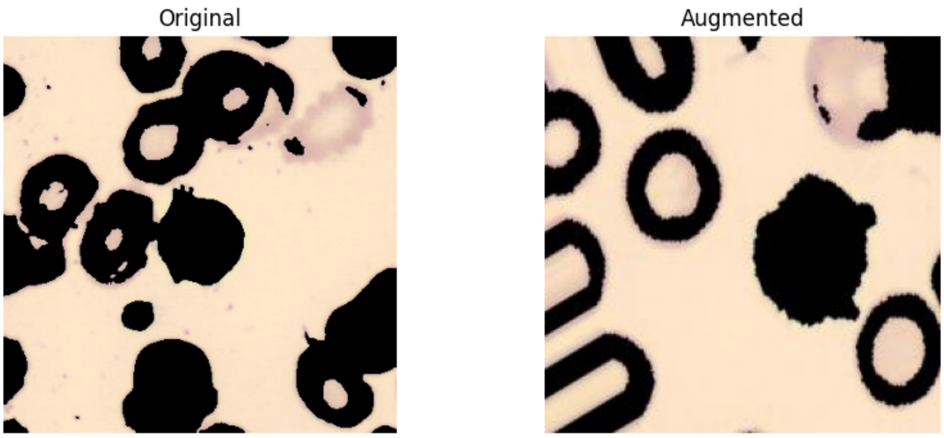
2.2.4 Distribution Analysis with Boxplots

The boxplot visualizations reveal substantial variability and extensive outlier patterns. MYO exhibits an exceptionally wide brightness range with outliers extending from approximately 115 to 140, indicating severe imaging inconsistencies. KSC, LYT and MON also display considerable spread with low-intensity outliers. The contrast distribution shows that LYT has an extremely broad range spanning from approximately 30 to over 110. For the RGB channels, all three exhibit extensive outlier populations. The prevalence and severity of outliers is substantially greater than in the first dataset and suggests the need for quality control measures before modeling.

The analysis reveals that this dataset exhibits significantly greater heterogeneity and imaging variability compared to the first dataset, with several classes (particularly KSC, MYO and PMO) showing extensive outlier populations. The extreme outliers observed - especially in MYO with brightness values ranging from 115 to over 210 - may indicate potential data quality issues.

3 Data Preprocessing

Before training machine learning models, several preprocessing steps were applied to ensure consistency and improve model performance. First, all images were resized to 128×128 pixel to standardize the input dimensions across the dataset. Standardization simplifies model training and reduces computational complexity.
Next, segmentation techniques were applied to convert images to grayscale, reduce noise using Gaussian blur, and isolate cells from the background using thresholding and morphological operations. These operations help remove background noise and highlight relevant cellular structures.
To improve dataset balance and prevent model bias toward dominant classes, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustments were applied to underrepresented classes to ensure class balance. Accordingly, dataset diversity increased and overfitting reduced. After augmentation, the dataset size increased to 28000 images and this larger dataset improves model stability and robustness. The following figure shows an example of an image before and after augmentation.
Finally, the dataset was divided into training and testing sets using stratified sampling, with 80% for training and 20% for testing. This ensured balanced representation of each class across both sets.



4 Feature Extraction

Since classical machine learning algorithms require structured input data, meaningful features were extracted from the images. In total, 51 features were extracted to capture morphological, color, and texture characteristics of the cells.

4.1 Extracted Feature Categories

The extracted features were grouped into several categories:

Shape Features: These features describe the geometric properties of the cell such as circularity, convexity, and solidity.

Color Features: Color features capture the staining characteristics of the cells using RGB values, such as the mean and standard deviation of the RGB values.

Texture Features: contrast, correlation, and smoothness.

All features were normalized using standardization before training and each image was represented as a feature vector, forming the input dataset for the classification models.

4.2 Features Importance

Feature importance indicates which features the model considers most useful for making predictions.
In machine learning, examining features importance helps us understand how the model "makes decisions", and it can also show which features we might not need. For example, in tree-based models like Random Forest, a feature is considered important if it is frequently used to split the data in a way that improves accuracy.
By checking features importance, we can focus on the most helpful features, simplify the model and gain better insight into what truly matters in the data.

5 Machine Learning Models

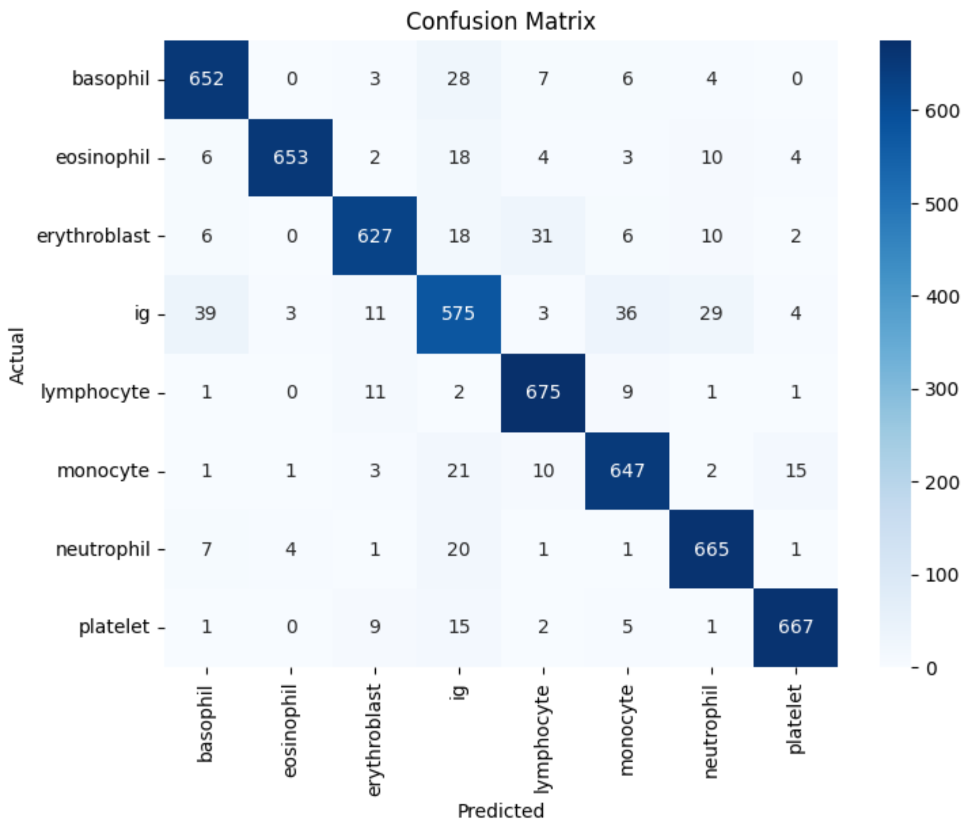
Supervised machine learning models were implemented and evaluated for the classification of the eight blood cell types of Mendeley blood cell dataset.

5.1 Random Forest

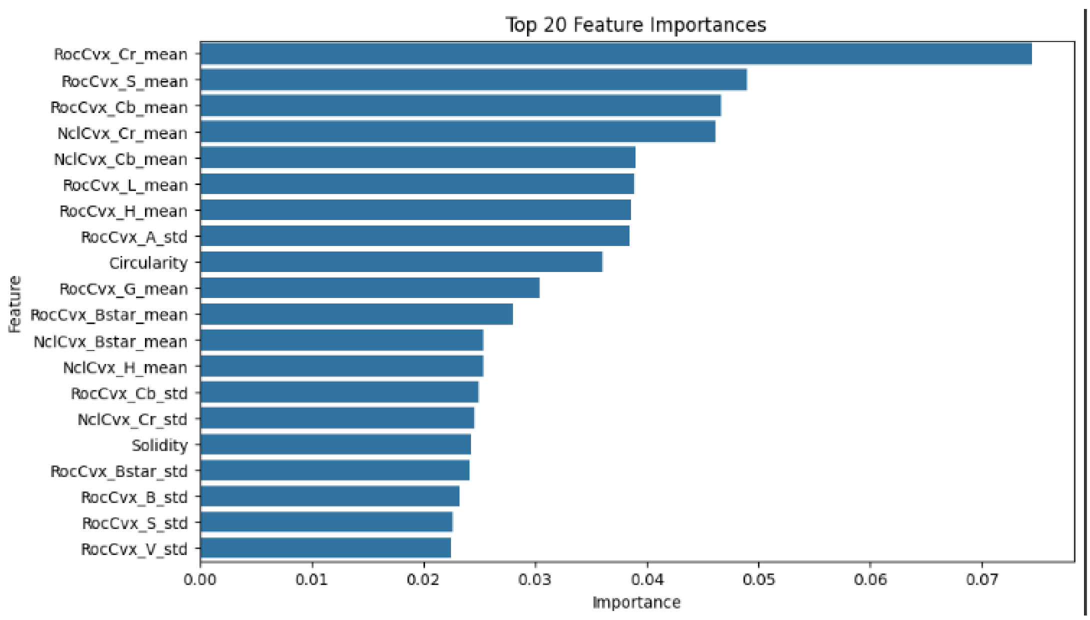
Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting. Using the processed dataset of 28,000 images, distributed evenly across eight classes, with 3,500 images per class, the model was tested on a balanced dataset consisting of eight classes, with 700 images per class (total = 5,600 images).
As the table shows, the Random Forest model achieved an overall accuracy of 0.92. The model performed well across most classes. The macro average and weighted average precision, recall, and F1-score are 0.92, indicating consistent and balanced performance across all classes. However, the Immature Granulocyte (IG) class showed slightly lower performance.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Basophil	0.91	0.93	0.92	700
Eosinophil	0.99	0.93	0.96	700
Erythroblast	0.94	0.90	0.92	700
IG	0.82	0.82	0.82	700
Lymphocyte	0.92	0.96	0.94	700
Monocyte	0.91	0.92	0.92	700
Neutrophil	0.92	0.95	0.94	700
Platelet	0.96	0.95	0.96	700
Accuracy		0.9216		5600
Macro Avg	0.92	0.92	0.92	5600
Weighted Avg	0.92	0.92	0.92	5600

Furthermore, the confusion matrix below confirms the strong overall performance of the model, consistent with the 0.92 accuracy. Most classes are well classified, except for IG class, which remains the primary source of classification error.
The diagonal elements (correct predictions) are dominant across all classes, confirming strong classification performance.
While most classes show correct predictions above 650 samples, the Immature Granulocyte (IG) class shows comparatively lower correct predictions (575), indicating greater classification difficulty. Misclassifications primarily occur between morphologically similar cell types.



Next, we looked at feature importance and found that the top 10 features captured most of the predictive power. Using only these features slightly reduced accuracy to 0.89, but made the model simpler and faster.
We also removed highly correlated features(24 features) to reduce redundancy, which again simplified the model with minimal impact on performance with accuracy 0.913.



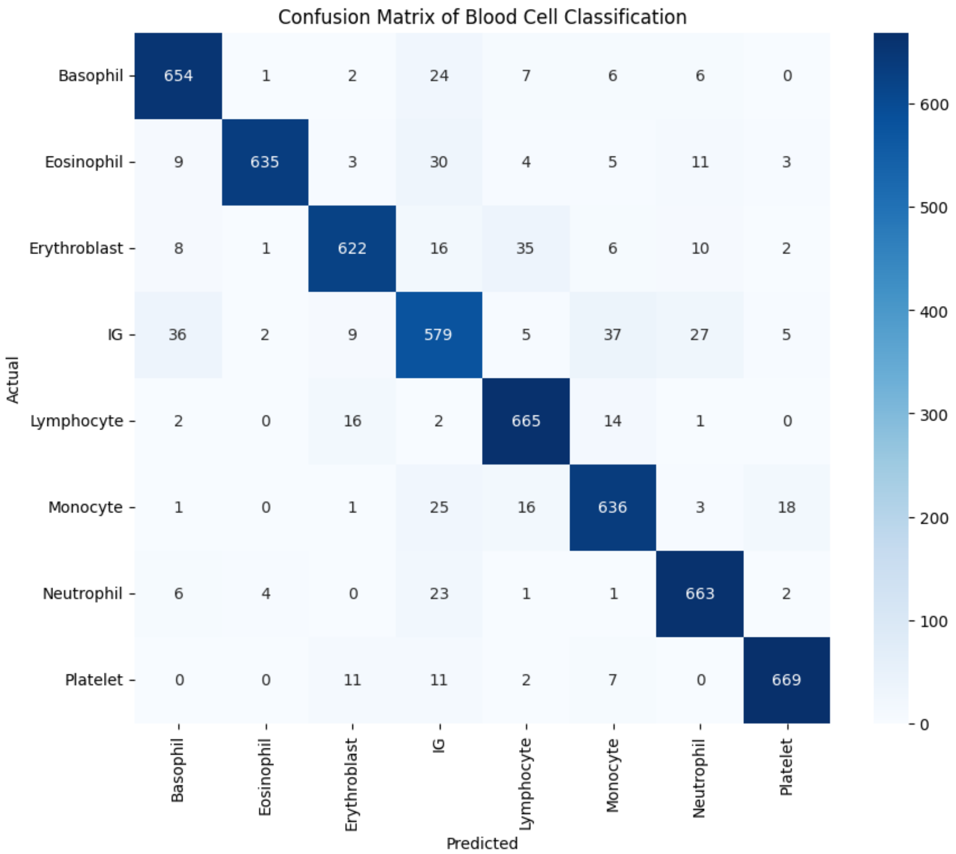
5.2 Tuned Random Forest

Hyperparameter tuning was performed using Grid Search, optimizing parameters such as the number of trees and the maximum tree depth. This further improved performance slightly, particularly in terms of the weighted F1 score. The confusion matrix shows that most misclassifications occurred between similar cell types, and also shows strong overall performance with 0.91 accuracy on a balanced dataset (700 samples per class). Most predictions lie on the diagonal, indicating reliable classification across cell types.

Best-performing classes: Platelet (669/700), Neutrophil (663/700), and Lymphocyte (665/700) show very high correct predictions with minimal confusion.
Most challenging class: IG (Immature Granulocyte) is the most confused class (579/700 correct), often misclassified as Monocyte, Basophil, and Neutrophil. Overall, the model performs well across most cell types, with errors mainly occurring between morphologically similar or immature cells.

In conclusion, the Random Forest model proved robust and effective for multi-class cell classification. Feature selection and hyperparameter tuning simplified the model and provided a clearer understanding of the most important features, while the overall accuracy remained high. Minor improvements could still be made for specific classes, like the IG cells, through additional feature engineering.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Basophil	0.91	0.93	0.92	700
Eosinophil	0.99	0.91	0.95	700
Erythroblast	0.94	0.89	0.91	700
IG	0.82	0.83	0.82	700
Lymphocyte	0.90	0.95	0.93	700
Monocyte	0.89	0.91	0.90	700
Neutrophil	0.92	0.95	0.93	700
Platelet	0.96	0.96	0.96	700
Accuracy		0.91		5600
Macro avg	0.92	0.91	0.91	5600
Weighted avg	0.92	0.91	0.91	5600



5.3 Support Vector Machine (SVM)

We developed and evaluated a Support Vector Machine (SVM) model with RBF kernel for multiclass classification using the same dataset of 28,000 images and 46 features.
To prepare the data, we implemented a preprocessing pipeline that handled missing values using mean imputation and standardized all numerical features with StandardScaler. Standardization was particularly important because SVM models are sensitive to differences in feature scale. Incorporating these steps into a pipeline also ensured that preprocessing was applied consistently during cross-validation, preventing data leakage.

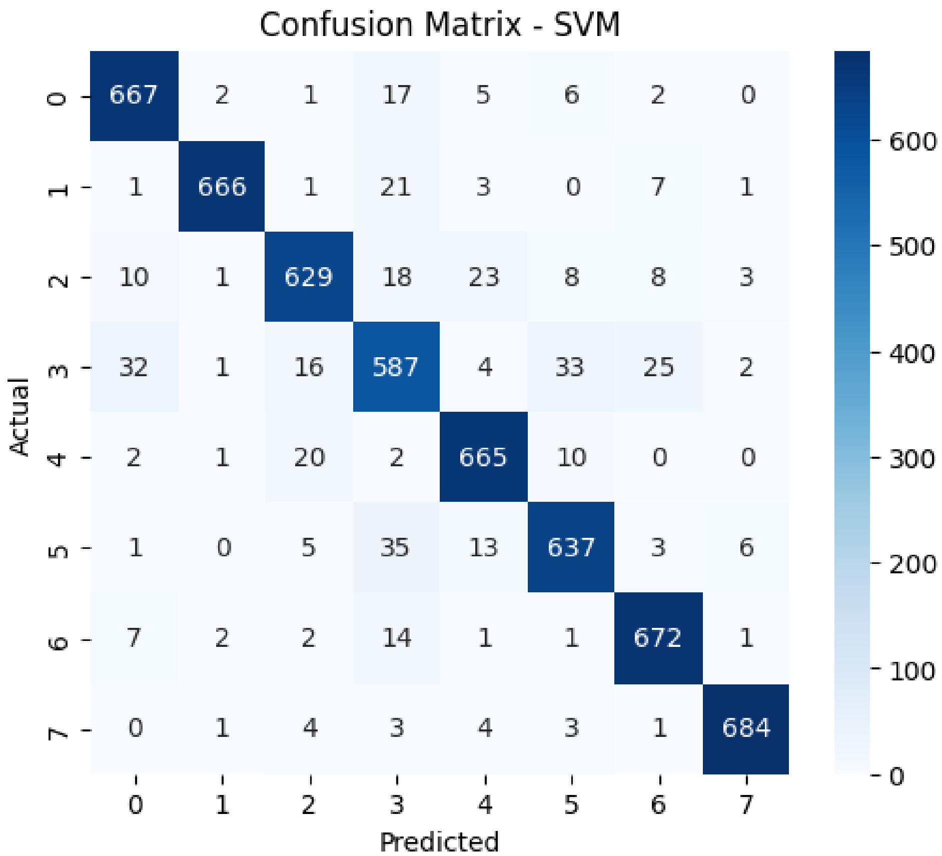
We first trained a baseline SVM model using RBF kernel with parameters C = 1 and gamma = ”scale”. This model achieved a test accuracy of 0.929 (approximately 0.93). Overall, performance was strong and consistent across most classes.

The classification report shows that the majority of classes achieved F1-scores above 0.93, indicating balanced precision and recall. However, class IG had comparatively lower performance, with: Precision: 0.84, Recall: 0.84, and F1-score: 0.84.

Additionally, the macro and weighted averages were both 0.93, with very little difference between them. This confirms that the dataset is well balanced and that the model performs consistently across all classes rather than being biased toward any specific class.
The confusion matrix of the baseline SVM confirms its strong overall performance. Most predictions lie along the diagonal, indicating correct classifications. Classes 6 and 7 (Neutrophil and Platelet) performed best, with 672 and 684 correct predictions out of 700, respectively.

In contrast, class 3 (IG) had the weakest performance, with 587 correct predictions. It was mainly misclassified as classes 0, 5, and 6 (Basophil, Monocyte, and Neutrophil). Some moderate confusion was also observed for classes 2 and 5 (Erythroblast and Monocyte).

Class	Precision	Recall	F1-score	Support
0	0.93	0.95	0.94	700
1	0.99	0.95	0.97	700
2	0.93	0.90	0.91	700
3	0.84	0.84	0.84	700
4	0.93	0.95	0.94	700
5	0.91	0.91	0.91	700
6	0.94	0.96	0.95	700
7	0.98	0.98	0.98	700
Accuracy		0.93		5600
Macro Avg	0.93	0.93	0.93	5600
Weighted Avg	0.93	0.93	0.93	5600



5.4 Tuned SVM

After hyperparameter tuning using GridSearchCV, the best parameters for the SVM were: C = 10, gamma = 0.1, and kernel = 'rbf'.
This tuned model achieved a test accuracy of 0.949, slightly improving over the baseline SVM (0.93).
Looking at the classification report, we found that the best performing classes: class Neutrophil: 0.99 F1-score, class Eosinophil: 0.97 F1-score, class Lymphocyte: 0.96 F1-score.
However, the lowest performing class was the IG class with 0.88 F1-score, indicating that some confusion still exists with this class.

Most other classes performed consistently well, with precision, recall, and F1-scores above 0.94. We observe that tuning increased overall accuracy from 0.93 to 0.95.

Macro and weighted averages for precision, recall, and F1-score all reached 0.95, indicating balanced performance across classes. Class IG remains the most challenging, similar to the baseline model, suggesting that its features overlap with other classes.

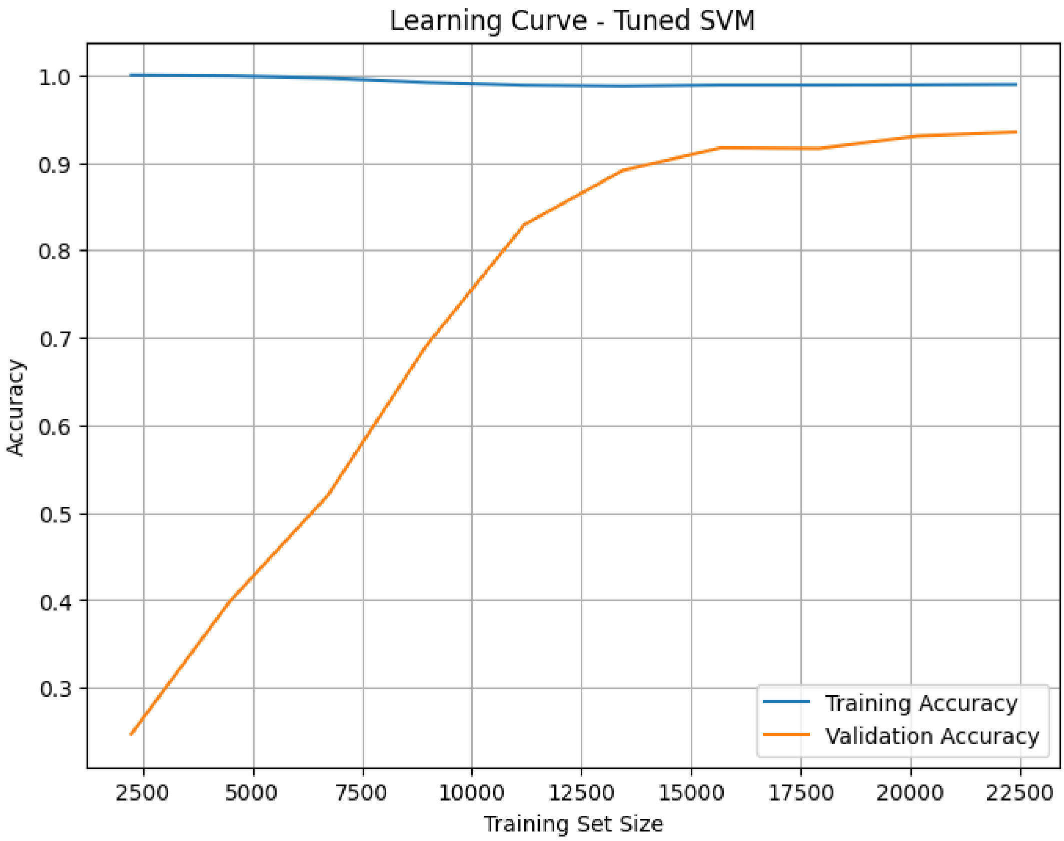
Classes Eosinophil, Lymphocyte, Neutrophil, and platelet show near-perfect classification, demonstrating that the model captures their feature patterns very effectively.
Moreover, increasing C to 10 increased the effectiveness of hyperparameter tuning, and using gamma = 0.1 with the RBF kernel gives the model flexibility to separate closely spaced classes without overfitting. We conclude that the tuned SVM shows strong and balanced performance across all classes, with a notable improvement in overall accuracy.
While Class IG still requires attention, the model is now highly reliable for this 8-class classification task. Future work could explore feature engineering or class-specific augmentation to further reduce confusion in Class IG.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Basophil	0.95	0.96	0.96	700
Eosinophil	0.98	0.96	0.97	700
Erythroblast	0.95	0.94	0.94	700
IG	0.89	0.87	0.88	700
Lymphocyte	0.95	0.98	0.96	700
Monocyte	0.94	0.95	0.95	700
Neutrophil	0.95	0.96	0.95	700
Platelet	0.99	0.98	0.99	700
Accuracy		0.95		5600
Macro Avg	0.95	0.95	0.95	5600
Weighted Avg	0.95	0.95	0.95	5600

5.4.1 Learning Curve Analysis – Tuned SVM

The learning curve shows that the tuned SVM starts by overfitting when the training dataset is small, and training accuracy is nearly perfect, but the validation accuracy is very low.
As more data are added, the validation performance steadily improves and eventually stabilizes around (0.93–0.94).

This means that with enough data, the model generalizes well. There is still a small gap between training and validation accuracy, suggesting a bit of overfitting, but overall the performance is stable.
Further improvements would likely require better features, stronger regularization, or experimenting with different kernels.
Briefly, with small datasets, the tuned SVM tend to overfit, but as training size grows, validation accuracy increases and stabilizes around (0.93–0.94), showing good generalization with sufficient data.



Model	Accuracy	Macro Avg F1	Best Parameters
SVM (Baseline)	0.93	0.93	Default
SVM (Tuned)	0.95	0.95	{C=10, gamma=0.1, kernel=rbf}
Random Forest (Baseline)	0.92	0.92	Default
Random Forest (Tuned)	0.91	0.91	{max_depth=None, n_estimators=200}

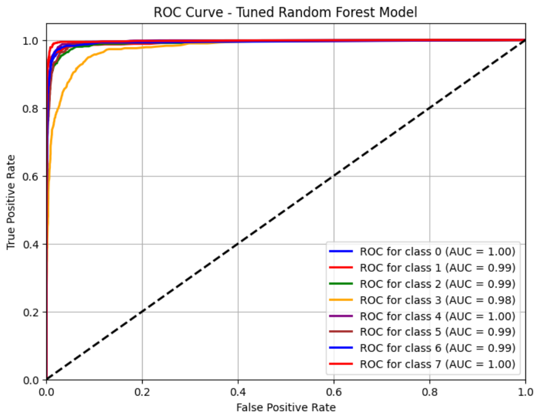
Table 1: Performance comparison of baseline and tuned machine learning models

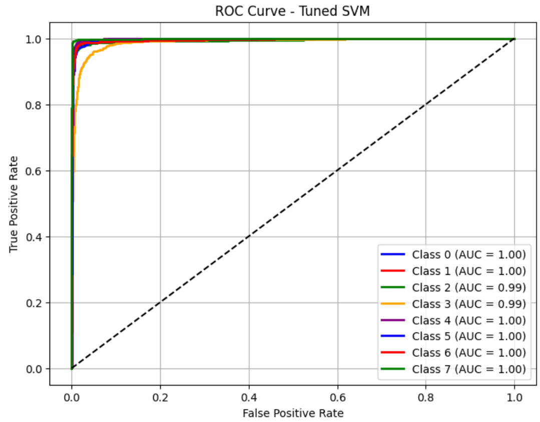
6 ROC Curve Analysis and Model Comparison

Both the tuned Random Forest and tuned SVM models demonstrate excellent classification performance. Most ROC curves are close to the top-left corner, indicating high true positive rates and low false positive rates. The AUC values for most classes exceed 0.99 or 1.00, showing strong class separability, while the Platelet class achieves perfect classification (AUC = 1.00) in both models.

For the tuned Random Forest, several classes such as Neutrophil, Lymphocyte, and Eosinophil achieve AUC values close to 0.98–1.00, indicating very strong performance. However, the Immature Granulocyte (IG) class shows a lower ROC curve, suggesting reduced separability and higher misclassification, which is consistent with the confusion matrix results.

Similarly, the tuned SVM model shows smooth and consistently high ROC curves across most classes, with many AUC values around 0.99 or 1.00 Despite this strong overall performance, the IG class again has the lowest AUC, confirming that it remains the most challenging class to classify.





Aspect	**Tuned Random Forest**	**Tuned SVM**
Overall AUC	Very high	Slightly higher consistency
Platelet detection	Perfect	Perfect
Strong classes	Neutrophil, Lymphocyte, Eosinophil	Same
Weakest class	IG	IG
Curve smoothness	Slightly more variation	More stable

7 Autoencoder Model

A convolutional Autoencoder, a type of artificial neural network, was used to learn the underlying structure of blood cell images and support anomaly detection. Autoencoders are unsupervised deep learning models that learn compact representations of data by reconstructing the input images. The model consists of two main parts: an encoder, which compresses the input image into a lower-dimensional representation, and a decoder, which reconstructs the image from this compressed representation. The network is trained by minimizing the difference between the original image and the reconstructed output.

Compared with traditional machine learning methods that rely on manually designed features, the autoencoder automatically learns complex morphological characteristics of blood cells directly from the images.
For this model, we used the Mendeley blood cell dataset, which contains 17,092 normal blood cell images. All images were resized to 128×128 pixels and normalized to the range [0,1] to ensure stable training. The dataset was divided into 80% training, 10% validation, and 10% testing.

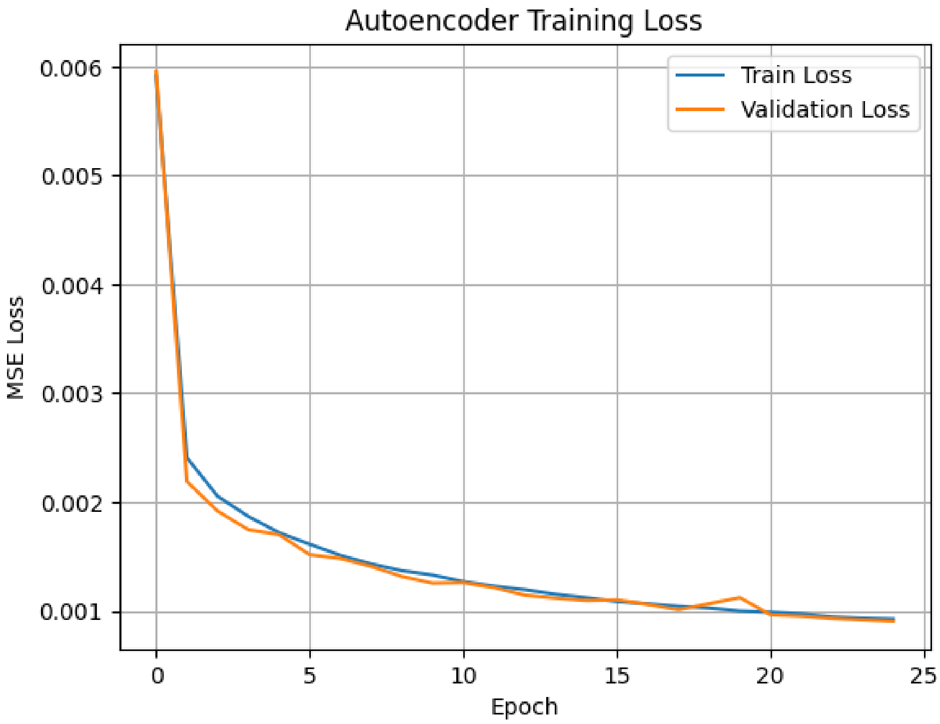
After training, the model achieved a final test loss of approximately 0.00088, indicating that the autoencoder was able to reconstruct the blood cell images with very small reconstruction error. The detailed training and evaluation results are summarized in the results table, which shows the performance of the model in the training, validation, and test datasets.

For anomaly detection, the reconstruction error was calculated using the validation set. We determined a threshold of approximately 0.0017. Images with reconstruction errors higher than this threshold are considered anomalous, while those below the threshold are classified as normal.

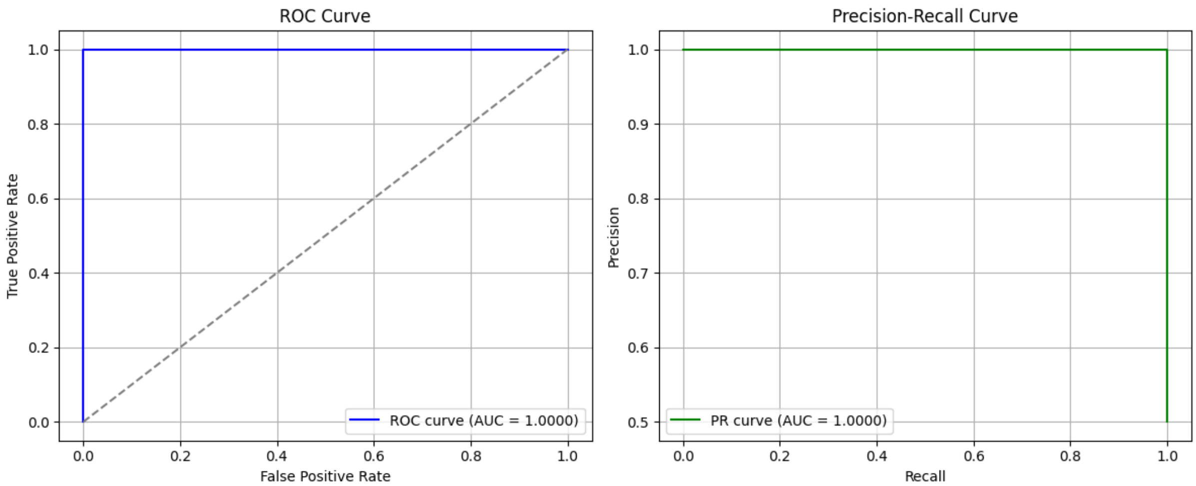
Overall, the low reconstruction loss and the results presented in the table indicate that the autoencoder successfully learned the normal patterns of blood cell images and can effectively identify unusual samples based on reconstruction error.

Category	Parameter/ Metric	Value
Model Parameters	Total Trainable Non-trainable	334,851 334,403 448
Training Configuration	Optimizer Learning rate Loss function	Adam 1×10^{-4} Mean Squared Error (MSE)
Training Results	Final-training loss Final-validation loss	0.000884 0.000884
Reconstruction Error	Normal images (MSE) Anomalous images (MSE)	∼0.00092 ∼3.5
Anomaly Detection	Threshold Decision rule	0.001749 MSE > Threshold → Anomaly; MSE ≤ Threshold → Normal
Performance Metrics	ROC-AUC F1-score PR-AUC	1.000 0.6667 1.000

The autoencoder training loss curve, as shown in the next figure, illustrates the model's behavior during training, showing both training and validation losses across epochs. The training loss decreases rapidly in the initial epochs, indicating that the model quickly learned the key structures and features of the blood cell images. As training progresses, the losses gradually stabilize, and the small gap between training and validation curves suggests good generalization with minimal overfitting. The final reconstruction loss of approximately 0.00088 demonstrates that the autoencoder can accurately reconstruct the images. Overall, the smooth convergence confirms effective training and reliable image reconstruction.



In addition, the ROC curve and Precision–Recall curve further confirm the effectiveness of the model. The ROC curve shows perfectly performance, and the AUC value reaches 1.00, indicating excellent separation between normal and anomalous samples. Similarly, the Precision–Recall curve also shows near-perfect performance, suggesting that the model can detect anomalies with high reliability.



8 Conclusion

This project explored the use of machine learning techniques for automated blood cell classification using microscopic smear images. Several supervised learning models, including Random Forest and Support Vector Machines, were implemented and evaluated using morphological, color, and texture features extracted from the images.

The results show that the tuned Support Vector Machine achieved the best overall performance, reaching an accuracy of about 95%. Most blood cell classes were classified with high precision and recall. However, the Immature Granulocyte (IG) class remained the most difficult to classify, mainly because its morphological characteristics are similar to those of other leukocyte types.

In addition to the classification models, a convolutional Autoencoder was used to learn the normal structure of blood cell images and detect anomalies through reconstruction error. The autoencoder achieved a very low reconstruction loss (around 0.00088) and was able to clearly distinguish between normal and anomalous samples, as reflected by the high ROC-AUC and PR-AUC values.

Overall, the results suggest that combining traditional machine learning methods with deep learning approaches can be an effective strategy for automated blood cell analysis. Such systems could potentially assist hematologists by reducing manual workload and supporting faster and more consistent diagnostic decisions.

9 Future Work

Although the models achieved good performance in this project, there are several directions that could be explored in future work. One possible improvement is to use deep learning models such as Convolutional Neural Networks (CNNs) for direct image classification. Unlike classical machine learning models, CNNs can learn features directly from the images, which may improve the overall classification performance.

Another direction is to further develop the anomaly detection approach. More advanced models, such as Variational Autoencoders (VAEs), could be tested to improve the detection of abnormal or rare blood cell types.

Future work could also include using additional datasets with more pathological blood cell images. This would help evaluate how well the models perform on more complex clinical cases and improve their robustness.

Finally, it would be interesting to integrate the developed models into a practical diagnostic tool that could assist hematologists in laboratory environments. Such a system could help speed up the analysis process and support more consistent blood cell classification.